



Important Formulas & Points to Remember: Finding the Unknown

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Understanding Equations

An **equation** is a statement that two algebraic expressions are equal.

LHS = Left Hand Side

RHS = Right Hand Side

Equation holds when $LHS = RHS$

Balancing Principle

Golden Rule of Equations

Performing the **same operation** on both sides of an equation **keeps it balanced**.

Allowed operations on both sides:

Add the same number

Subtract the same number

Multiply by the same number

Divide by the same number (non-zero)

This principle is the foundation of equation solving.

Inverse Operation Rules

To isolate the unknown:

(a) Additive Inverse

If a term is **added or subtracted**, remove it by applying its inverse.

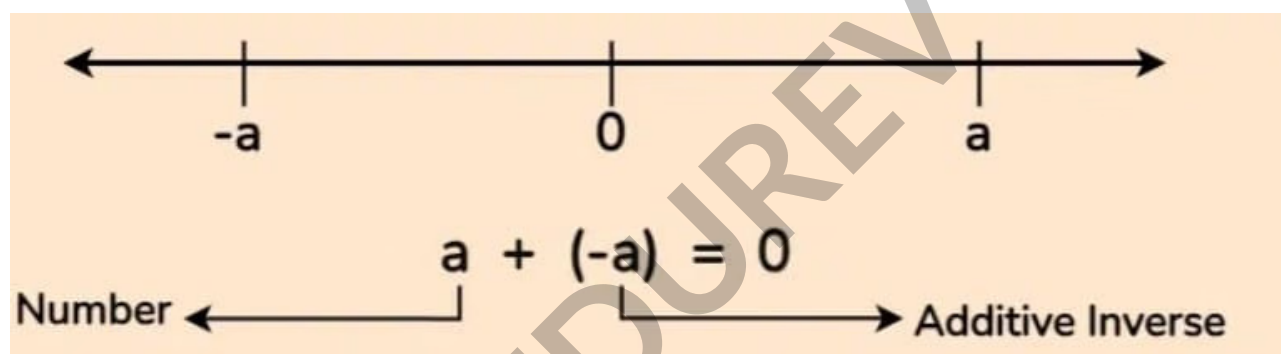
Examples of required operations:

Remove +7 → subtract 7

Remove -5 → add $-(-5)$ which is +5

General form:

$$Ax + B = C \rightarrow Ax = C - B$$



(b) Multiplicative Inverse

If the unknown is multiplied by a constant, divide both sides by that constant.

General rule:

$$Ax = C \Rightarrow x = \frac{C}{A}$$

(c) For equations of the form:

$$\frac{u}{k} = m$$

Multiply both sides by k:

$$u = m \times k$$

Steps to Solve Any Linear Equation

1. Move all x-terms to one side.
2. Move all constants to the other side.
3. Combine like terms.
4. Use inverse operations to isolate the unknown.
5. Check the answer by substituting back.

Equations With Unknowns on Both Sides

General equation:

$$Ax + B = Cx + D$$

Steps:

Bring like terms together: $Ax - Cx = D - B$

Simplify: $(A - C)x = D - B$

Solve:

$$x = (D - B) / (A - C)$$

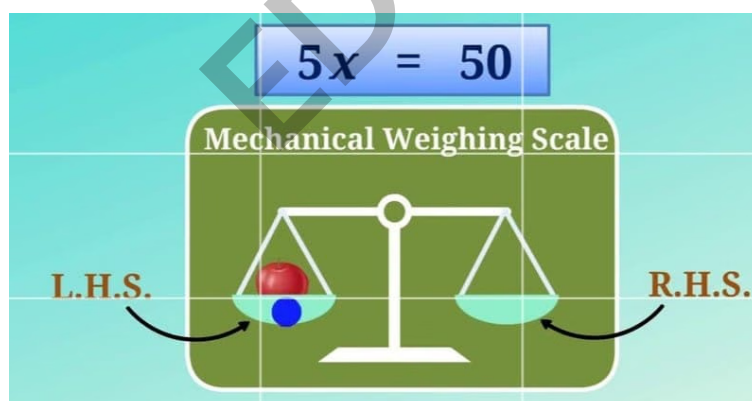
This direct formula is extremely useful.

Weighing Scale Method

If a balance has equal weights on both sides:

Removing equal items keeps it balanced

Adding equal items keeps it balanced



Same idea applies to algebra:

If $A + x = B + x$, then $A = B$.

Algebraic Properties

Distributive Property:

$$a(b + c) = ab + ac$$

Combining Like Terms:

Add/subtract coefficients of the same variable

Add/subtract constants separately

General Solving Forms

Form 1: Removing Constants

$$Ax + B = C$$

$$\rightarrow Ax = C - B$$

$$\rightarrow x = (C - B) / A$$

Form 2: Removing Coefficients

$$Ax = B$$

$$\rightarrow x = B / A$$

Form 3: Fractional Equations

$$x/a = b$$

$$\rightarrow x = ab$$

Form 4 (Unknown both sides): Direct Brahmagupta Formula

$$Ax + B = Cx + D$$

$$\rightarrow (A - C)x = D - B$$

$$\rightarrow x = (D - B) / (A - C)$$

How to Check Your Solution

Substitute the found value of x into the original equation.

Calculate LHS and RHS.

If both sides match, the solution is correct.